

 AES Mong Duong	AES Mong Duong Power Co., Ltd.	AMS Procedure No.	AES-MD- AMS-SP-009
	Asset Management System	Effective date	20 May 2025
	Continuous Improvement Procedure	Rev	2

AES MD

Continuous Improvement Procedure

AES-MD-AMS-SP-009	
Approved by: Johnny Tanis	

Review No.	Date	Prepared by	Verified by	Reviewed by
0	15 April 2019	Nguyen Huu Hung, Nguyen Thanh Binh	Nadessan Sandirasegaran	Nadessan Sandirasegaran
1	20 April 2020	Nguyen Thanh Binh	Nadessan Sandirasegaran	Nadessan Sandirasegaran
2	20 May 2025	Vu Thi Huyen Trang	Vu Thi Huyen Trang	Nguyen Duy Quang

The approved version on 20 May 2025 is the current valid one. All the previous versions of this procedure shall be automatically invalid

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REVISION HISTORY

No.	Date of revision	Description of change	Reviewed by	Approved by
1	20 April 2020	Change company's name and logo	Nadessan Sandirasegaran	Kevin Pierce
2	20 May 2025	MDII PEX award contest	Nguyen Duy Quang	Johnny Tanis
3				
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1. OBJECTIVE

This procedure describes the process to be used to identify, assess, prioritize, plan, execute and closeout Continuous Improvement (CI) projects. All CI projects are required to follow this process.

Key objectives of the process are to:

- Identify opportunities and perform assessing, prioritizing and implementing actions to achieve continuous improvement in the optimal combination of costs, asset-related risks and the performance and condition of assets and asset systems.
- Use a structured problem-solving process and associated techniques to address asset-related problems and performance gaps by finding the root cause.
- Actively seek and acquire knowledge about new Asset Management related technology and practices including new tools and techniques through innovation, technical excellence and development of sustainable practices.
- Evaluate the Continuous Improvement projects to establish their potential benefit to the company.

2. SCOPE

Continuous Improvement Procedure shall cover the performance and condition of assets and asset systems across the entire life cycle. The project management process involves implementing actions aimed at planning, executing and controlling various activities to achieve specific goals and standards.

The process applies to both day-to-day problems solving and more substantial improvement projects.

3. DEFINITIONS

- **Project** - A temporary endeavor undertaken to create a unique product, service or result. The project should have clear goals, outcome measures, start and end dates that meet negotiated and explicit stakeholders.
- **Project Leader** - The person assigned to lead the team that is responsible for achieving the project objectives and the final delivery of products and/or services generated.

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- **PDCA cycle** - An iterative four-step management method used for the control and continuous improvement of processes and products. PDCA stands for Plan Do Check Act.
- **DMAIC Six Sigma** - A data-driven improvement cycle used for improving, optimizing and stabilizing business processes. DMAIC stands for Define Measure Analyze Improve Control.
- **Project Charter** - A document issued by the project initiator that formally authorizes the existence of a project and provides the project leader with the authority to apply organizational resources to project activities.
- **Key Performance Indicators (KPIs)** - A type of performance measurement to evaluate the success of a particular activity in which it is engaged.
- **SIPOC diagram** - A tool used by a team to identify all relevant elements of a process improvement project before work begins. SIPOC stands for Suppliers Inputs Process Outputs Customers.
- **Business Case** - A tool to support decision making on whether or not to run the project, depending on its financial viability. Provides information on the financial impact and strategic issues relating to the project.
- **Stakeholders** - Individuals or groups whose interests may be affected by the results obtained during the execution and completion of the project.
- **Action plan** - Detailed plan outlining actions needed to reach one or more goals.
- **Project Schedule** - A plan for executing the project that presents linked activities with planned dates, durations, milestones and resources.
- **Mong Duong II Performance Excellence (MDII PEX)** - The transformation vehicle that enables the organization to create a competitive advantage and deliver results through continuous improvement, knowledge sharing and innovation.

4. ROLES AND RESPONSIBILITIES

Department Managers - shall be:

- Accountable and responsible for aligning continuous improvement activities with Asset Management strategic plans;

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- Accountable for conducting cross-functional daily meetings to discuss asset performance issues;
- Accountable and responsible for conducting regular planning meetings to identify and solve tactical and strategic performance problems;
- Accountable and responsible for identifying business data requirements to implement Asset Management plans and objectives;
- Accountable for implementing methods to capture and store required data;
- Accountable and responsible for identifying continuous improvement project list needs/opportunities;
- Accountable and responsible for performing cost-benefit analyses for identified projects;
- Accountable and responsible for establishing continuous improvement project list for coming years and identifying scope and leaders;
- Accountable and responsible for executing continuous improvement projects, evaluating and reporting project results;
- Accountable for incorporating project lessons learned into plans, training and Standard Operating Procedures (SOP);
- Accountable and responsible for identifying and tracking project KPIs against targets, utilizing to measure continuous improvement progress.

Area Leaders - The Area Leaders shall be:

- Responsible for conducting cross-functional daily meetings to discuss asset performance issues;
- Responsible for conducting regular planning meetings to identify and solve tactical and strategic performance problems;
- Accountable for implementing methods to capture and store required data;

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- Responsible for identifying continuous improvement project list needs/opportunities;
- Responsible for incorporating project lessons learned into plans, training and Standard Operating Procedures (SOP);
- Responsible for identifying and tracking project KPIs against targets, utilizing to measure continuous improvement progress.

Project Engineers (Engineering Department) - Project Engineers shall be responsible to coordinate continuous improvement projects, evaluate and report project results.

MDII PEX Champion – MDII PEX Champion shall coordinate the MDII PEX program implementation and provide guidance to the MDII PEX Project Leaders; schedule quarterly MDII PEX Consolidation project status reviews with the MDII PEX Teams.

Project Sponsor - Usually business area executive or process owner who has vested interest in outcome of team results. Project Sponsor approves team formation and project charter; participates in the team startup; establishes guidelines for solution path; meets with team at regular intervals; provides guidance and review key work products; ensures team has adequate resources; informs executives in related business areas; communicates team results to organization. The Project Sponsor is a primary approver of team recommendations.

Project Leader - Project Leader shall provide leadership and manage overall effort of the team; set meeting agendas; monitor work progress and review quality; resolve team schedule and quality problems; take assignments and perform work; manage project issue resolution process; keep the sponsor informed of team progress.

Project Team Member - Project Team Member shall provide knowledge and expertise; perform the work of the project; take assignments and work collaboratively; gather and analyze data; develop work products in timely and quality manner; actively participate in group discussions; keep the team leader informed of task progress and issues.

Asset Management Champion - Asset Management Champion shall be responsible for identifying and training personnel to ensure people are competent in the process and implementation of Continuous Improvement (CI);

5. PROCESS

5.1.Process Description

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Stage 1 Initial Phase

Identify current challenges and opportunities for improvement: It consists of identifying opportunities to improve an existing process and convert them into projects. This identification can be done by any Mong Duong II employee or third-party consultants by analysing the Key Performance Indicators (KPIs) or/and the Financial Plan noting opportunities and detachments between years.

Designate Project Leader: Department Manager shall assign a professional who has the knowledge, ability and willingness to lead the project implementation.

Identify methodologies to be applied: At this stage, the Project Leader shall identify the management methodologies that will be applied to the project. The methodology of implementation of Continuous Improvement Projects shall be set according to the type of problem. For problems of unknown cause and low complexity the PDCA methods and tools shall be used. For problems of unknown cause and medium or high complexity the DMAIC (Six Sigma) methods and tools shall be applied. For problems of known cause, a business case shall be used.

Develop Project Charter: This project activity covers the initial analysis of the project necessity characteristics:

- Problem Statement.
- Objectives to be achieved.
- Project Key Performance Indicators.
- Business Case - the initial estimate of the financial benefits that will be obtained.
- Preliminary estimate of the project schedule.
- Project Team with multi-skilled and cross-functional membership.
- Process mapping - SIPOC diagram, flowchart, etc.
- Stakeholders - the parties interested in the project such as functional areas, professionals themselves and/or others, regulatory and public agencies, clients, consultants, contractors, business partners and among others, that somehow might influence or be influenced by the outcome of the project.

Approve Project Charter: The Project Charter shall be approved by the Project Sponsor to start the planning phase.

Stage 2 Planning Phase

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Define the Problem: The purpose of this stage is to develop full understanding of the assignment and refine the problem statement. As a result, a properly crafted Problem Statement and a common understanding of the problem among team members should be established.

Describe the Current Stage: Project team members gather as much as possible data and information that helps to verify the problem and issues faced and to fully understand the nature of the problem and its impacts on stakeholders. As a result, documented, fact-based understanding of the current state of the problem, indicators of performance, process analysis identifying cost drivers, issues, etc. shall be created.

Identify the Root Causes: The objective is to identify and prioritize the root causes of the problem and validate them based on data collection and analysis as per “STD0009 Root Cause Analysis”.

Elaborate Action Plan, Project Schedule and Project Budget: As the project progresses and the root causes are identified, the Project Leader shall create a detailed action plan to address each cause. After the preparation of the action plan the Project Leader shall develop a detailed schedule for implementing the Project. Control plans and activities are designed as well.

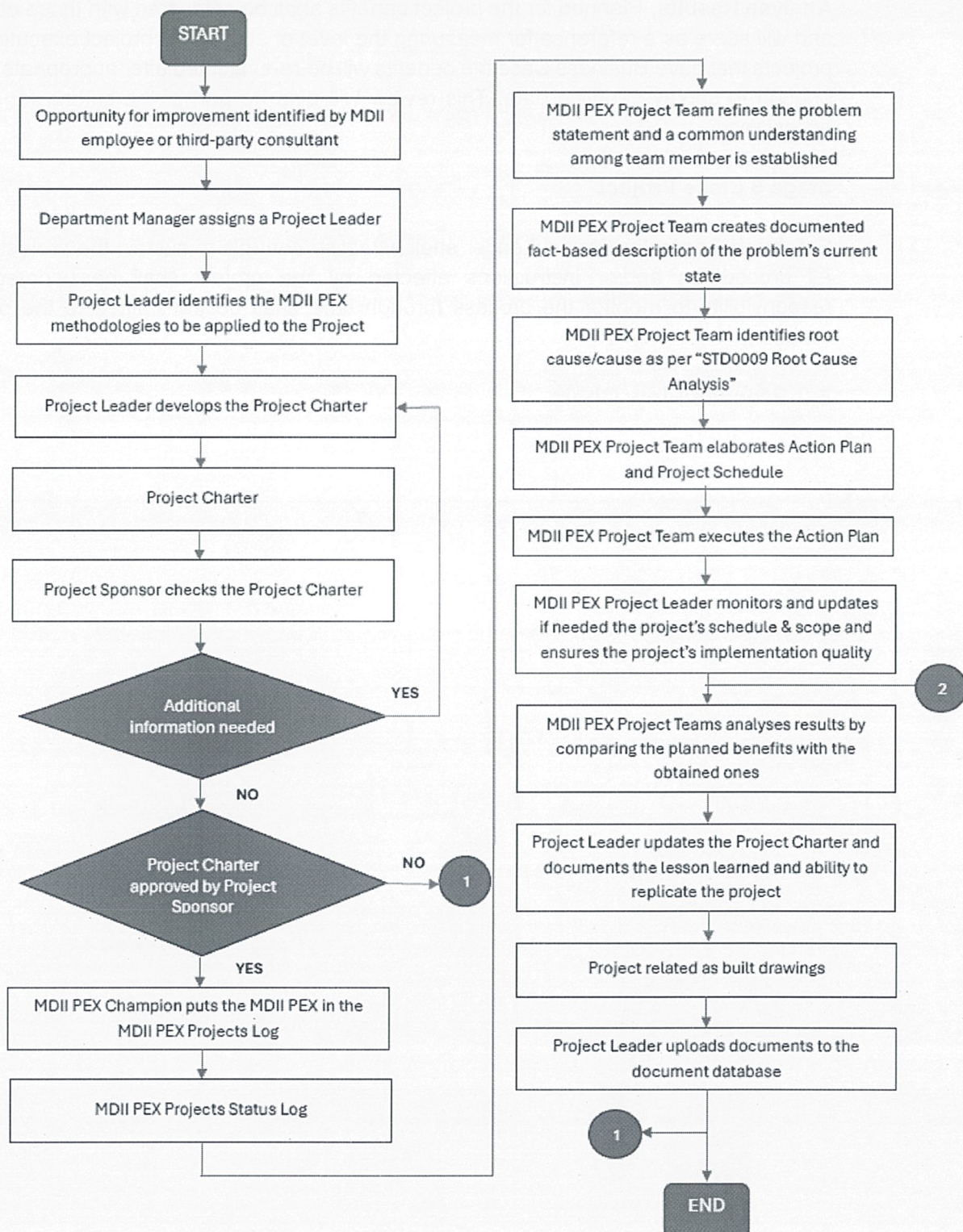
Stage 3 Execution Phase

Execute Project Plan: Implementation of project activities according to the methodology of implementation defined in the action plan and project schedule. If needed depending on the project implementation cost the Capital Expenditure Procedure based on “STD0002 Capital and Operating Expenditures” or Management of Change Procedure based on “STD0003 Management of Change” shall apply. In these cases, the relevant procedure shall be used for project execution management.

Track Project: Project Leader shall monitor and update the Project Schedule and will be responsible for reporting progress to the Project Sponsor, possible deviations of execution time and set schedule recovery plan in case of delays. Project Leader shall manage the project scope is being fulfilled and whether changes to the original scope are required. If there is a need for scope change the impacts in term, scope, quality and project budget shall be analysed and submitted to the Project Sponsor for approval. After endorsement of the change, it will be the Project Leader to update all project documentation. The project will be constantly monitored by the Project Leader to ensure its quality and compliance with the requirements.

Stage 4 Check Phase

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5.3. Process Control

MDII PEX Projects Log Review - The MDII PEX Champion and MDII PEX Teams shall review project status quarterly. The purpose of the review is to monitor status of the projects, ensure alignment of all improvement activities with their overall Asset Management objectives; asset-related losses are understood and broadly quantified; short- and medium-term improvement focus areas and initiatives are defined.

Audits - The Asset Management Champion shall lead to auditing the Continuous Improvement process and reporting any findings at least annually.

Training – The Asset Management Champion shall ensure that each member of Continuous Improvement Project Team should have adequate training in the MDII PEX methodology.

6. MDII PEX AWARD CONTEST

6.1. Participants

All employees of the AES Mong Duong Power Company. Group participation is allowed, with a maximum of 5 members per group.

6.2. Contest Categories

The MDII PEX Innovation Award Contest encourages participants to propose innovative ideas across various categories, including Technical Innovations, Safety and Environmental Enhancements, Digital Transformation, as well as Finance, Commercial, and other Management Improvements.

6.3. Submission Requirements

- Registration form with details of participants.
- A project proposal including:
 - Project title and category.
 - Problem statement and proposed solution.
 - Implementation plan and expected outcomes.
 - Cost-benefit analysis.
 - Supporting documents (diagrams, calculations, prototypes if available).

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6.4. Judging Criteria

- Methodology and Tools: How the idea is developed and applied.
- Financial Benefits: Cost savings or revenue generation.
- Intangible Benefits: Improvements in safety, reputation, or morale.
- Strategic Contribution: Aligns with company goals and values.
- Creativity/Innovation (Bonus): Unique and creative approach.
- Presentation (Bonus): Clarity and quality of presentation.
- Replication Potential (Bonus): Can the idea be applied elsewhere?
- Judge's Pick: Personal preference of the judging panel.

6.5. Evaluation Process

The evaluation of entries in the MDII PEX Competition is conducted by a designated Judging Panel. This panel performs a thorough assessment and scoring of each submission based on pre-defined criteria.

- Composition of the Judging Panel:
 - Plant Managers
 - Department Managers
- Responsibilities of the Judging Panel:
 - Review all submitted entries in detail.
 - Evaluate each submission based on the competition's evaluation criteria.
 - Assign scores and provide feedback where applicable.
 - Ensure transparency, fairness, and consistency throughout the evaluation process.

6.6. Timelines

The MDII PEX competition is held annually and is divided into two distinct stages: the registration period and the competition phase.

- Registration Period:
 - The registration period starts in May of each calendar year and remains open throughout the year.
 - Teams are encouraged to submit their registration to the MDII PEX champion each quarter.
 - The final deadline for registration is in February of the following year.

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- On going review: The MDII PEX Champion shall schedule quarterly MDII PEX Consolidation Status report reviews with the MDII PEX Teams.

- **Competition Phase:**

The competition itself takes place after the registration period, in April of the following year. Registered teams present their projects for evaluation and scoring by the Judging Panel.

6.7. Awards and Recognition

- First Prize: \$800 and a Certificate of Excellence
- Second Prize: \$400 and a Certificate of Achievement
- Third Prize: \$300 and a Certificate of Achievement

6.8. Implementation of Winning Ideas

- Support and funding will be provided to develop and integrate feasible innovations.
- Recognition in the plant's newsletter and company website.

6.9. Rules and Regulations

- All entries must be original and must not violate intellectual property rights.
- Proposals must be submitted in English and follow the specified format.
- The decisions of the judging panel are final and not subject to appeal.
- Participants must ensure that ideas are safe, practical, and cost-effective.

7. DOCUMENTED INFORMATION

MDII PEX Project Charter (see Attachment 1) shall be used for identifying the problem statement and objectives, Key Performance Indicators (KPIs) to be improved, team members, project schedule and plan to identify clear deliverables. The charter shall describe the PDCA (DMAIC Six Sigma) methods and tools applied as well.

MDII PEX Project Status Report (see Attachment 2) shall be used for consolidating all projects' relevant data and maintained by the MDII PEX Champion on a quarterly basis.

MDII PEX Project Submission Templates (see Attachment 3) shall be utilized for Registration of entries for the MDII PEX contest.

MDII PEX MDII PEX Judges Scoring - Screening Criteria (see Attachment 4) shall be used for assessment and scoring of Judges Panel.

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The Records shall be kept on Mong Duong II file server for reference.

8. REPORTING

The MDII PEX Champion shall be responsible for preparing and distributing a Quarterly MDII PEX Projects Status Report. The report is distributed to all members of the MDII PEX Teams by e-mail.

All reports shall be uploaded to the document data base project wise.

9. REFERENCES

9.1. Internal (Asset Management Standards)

STD0001: Asset Management System
STD0002: Capital and Operating Expenditures
STD0003: Management of Change
STD0005: Asset Operation and Maintenance
STD0008: Training Awareness and Competence
STD0009: Root Cause Analysis
STD0010: Performance Monitoring

9.2. External

BS ISO 55000:2014: Overview, principles and terminology
BS ISO 55001:2014: Requirements of management systems
BS ISO 55002:2014: Guidelines for the Application of BS ISO 55001

10. ATTACHMENTS

Attachment 1: MDII PEX Project Charter
Attachment 2: MDII PEX Project Status Report
Attachment 3: MDII PEX Project Submission Templates
Attachment 4: MDII PEX Judges Scoring - Screening Criteria

COMPANY

PROJECT TITLE

Problem/Issue Statement

OBJECTIVE

KPI

K Ind	KPI	Unit	Current Va	Goal
Result				
Effort				
Effort				

PROJECT TEAM

Stage	Start	End	Days	Diliverables	Responsible
PLAN					
Define the problem					
current state analysis					
Find root causes					
Plan Solution					
Do					
Check					
Act					

Plan										
Define the Problem										
Current State Analysis										
Find Root Cause										
Plan Solution										
Do										
Check										
Act										
	1-Jan	2-Jan	3-Jan	4-Jan	5-Jan	6-Jan	7-Jan	8-Jan	9-Jan	10-Jan

MDII PEX PPROJECT STATUS REPORT

No.	Project Description	Project Lead	Project Sponsor	Initial investment (USD)	Expected Benefit/ Year (USD)	Target Schedule		% Complete (Actual) as of Month
						Start	Finish	

Attachment 3
MDII PEX SUBMISSION - WHITE PAPER

PROJECT NAME: Put Project Title Here

PROJECT TYPE: MDII PEX or Replication

PROJECT TEAM: Name, Name, Name

LOCATION: AES Business, SBU

ABSTRACT

Provide a 150 word or less summary of your project

BUSINESS CASE

Discuss the challenge or issue addressed or opportunity for improvement and provide context to the issue or opportunity. Be sure to include overall project goals and objectives.

METHODOLOGIES

Discuss the MDII PEX methodologies and tools used to address the issue or opportunity

IMPLEMENTED SOLUTIONS

Discuss the solution and the process used for implementation

RESULTS

Review the results and outcomes of the implemented solutions. Be sure to include the project's impact on business or group KPIs.

PEOPLE ENGAGEMENT

Discuss the level of teamwork and engagement of people from across functions and organizational levels. Include other engagement activities according to judging criteria guidelines.

CONCLUSION

From a replication perspective, please review lessons learned, how this project can be replicated by other AES businesses and next steps for replication plan.

TEAM PHOTO

Insert a group photo and include name and title.

Note: Please ensure that project summary has enough detail to address the subcomponents of the main judging criteria. Also, the summary should include definitions for all technical terms to enable those unfamiliar with the project to understand the full content.

MDII PEX Project Judging

Department Manager

NAME

Department Manager

1

2

3

4

5

6

7

Judging Criteria	Scoring Options	Score	Project Name	Project Name	Project Name	Project Name	Project Name	Project Name
Methodology and Tools	There is no consistent application of a methodology (Insufficient)	1						
	Methodology is consistently applied, and the project exhibits clear KPIs, metrics, an adequate problem statement, business case, and quantifiable objectives (Sufficient)	2						
	Meets all previous requirements and demonstrates a coherent and sequential use of APEX tools, following a logical order (Good)	3						
	Meets all previous requirements, incorporates descriptive statistics (i.e. mean, median, range) for a more in-depth analysis, and defines some actions to sustain improvements (Very good)	4						
	Meets all previous requirements and establishes a process that ensures sustained improvements (Excellent)	5						
Financial Benefits	\$1 to \$100,000	1						
	\$100,000 to \$1,000,000	2						
	More than \$1,000,000	3						
	None	0						
Intangible Benefits	Partly relevant with minimal impact	1						
	Relevant with noticeable positive impact	2						
	Highly relevant with transformative impact	3						
	Not replicable	0						
Replication Potential	Replicable to 1 location or business function	1						
	Replicable to multiple locations or business functions	2						
	Replicable globally	3						
	Somewhat aligned with values and business objectives	1						
Contribution to Strategy and Values	Clear actions taken that contribute values and business objectives	2						
	Takes values and business objectives to the next level	3						
	Creativity/innovation not apparent	0						
	Creativity/innovation evident in the solution or during project development	1						
Creativity/Innovation (Extra Point)	Average or Below Average Presentation	0						
	Excellent Presentation	1						
	Not a Best Project	0						
	3rd Best Project	1						
Judge's Pick	2nd Best Project	2						
	1st Best Project	3						
Notes								